



Morphological and molecular characterization of *Ophiotaenia tigrina* in *Hoplobatrachus tigerinus* of YSR Kadapa District, Andhra Pradesh, India

Hemalatha Mannela, Srinivasa Kalyan Chadamala, Anu prasanna Vankara*

Yogi Vemana University, Kadapa, Andhra Pradesh, India.

*For correspondence E-mail: dr.anu@yogivemanauniversity.ac.in

Article Info: Received 17 Aug 2019; Revised: 27 Nov 2019; Accepted: 29 Nov 2019

ABSTRACT

Ophiotaenia tigrina Woodland, 1925 is a proteocephalid cestode (endoparasite) isolated from the small intestine of *Hoplobatrachus tigerinus* of YSR Kadapa District, Andhra Pradesh during February 2013 to February 2015. The main intention of the study was to evaluate the morphological condition and molecular characterization of *O.tigrina* by using 28S ribosomal DNA and to find out the phylogenetic position of this species with bioinformatics tools such as NCBI blast and Mega 6 softwares. The parasites obtained from the host samples of four sampling sites of YSR Kadapa district, Andhra Pradesh were preserved for morphological analysis, scanning electron microscopy studies and molecular characterization. The morphological data obtained from the whole mounts and the sequence of the cestode parasite obtained through molecular analysis was submitted to the NCBI Genbank (KY574290). A thorough comparison between the gene sequence of the present parasite and the available gene sequences of various cestode parasites from various vertebrate hosts in the NCBI Genbank was made to provide the accurate identification of the parasite. The pairwise alignment of gene sequences showed 70% similarities with the order Proteocephalidea. The study presents a small molecular phylogenetic analysis of the tapeworms of family Proteocephalidea and order Proteocephalidae. These types of analyses can reveal the relationship between different orders, families and species of the cestode parasites from the different hosts.

Keywords: *Ophiotaenia tigrina*, proteocephalid cestode, *Hoplobatrachus tigerinus*, molecular characterization, YSR Kadapa.

1. INTRODUCTION

Vertebrate host have been representative hosts in the study of the community and evolutionary ecology of parasites for the past few decades. Species

recognition is one of the central problems in vertebrate metazoan parasites. Parasites are generally minute, soft bodied and have few morphological characters. But, tapeworms (Platyhelminthes:

Cestoda) are a group of parasitic endoparasites of vertebrates that display a long ribbon shaped body form with wide variations in their life-cycles and host relations¹. Anuran amphibians serve as the most preferred vertebrate hosts for a number of metazoan endoparasites like digeneans, cestodes, nematodes, acanthocephalans and protozoans and ectoparasites such as monogeneans and leeches. Work on cestode parasites of anuran amphibians was on track way back in 18th Century itself, but the molecular work came into limelight in the recent years i.e., from 20th century onwards. Cestodes are long (10-20cm) and slender worms of whitish color and simply suitable for molecular analysis when compared with other metazoan parasites. Cestodes of the family Proteocephalidae Mola, 1928 are parasites of freshwater fishes, amphibians and reptiles²⁻⁵. It is a widely distributed family of cestodes with practically 300 documented species in 66 genera and 13 subfamilies. The morphology, host specificity and biological traits can be used to construct the phylogenetic relationships of proteocephalid cestodes but it is still intricate to come up with a reliable conclusion on the phylogenetic relationship between the members of proteocephalidae. Recently, many studies are being focused to show high level of interspecific variations for molecular characters and these variations are helpful for characterization of species in *Proteocephalinae* subfamily⁶. *Ophiotaenia* La Rue, 1911 is one among the 19 genera within the subfamily Proteocephalinae with nearly 96 species parasitizing amphibians, reptiles and birds and exclusively 25 species among them parasitize amphibians^{5,7-10}. The genus *ophiotaenia* was first described by La Rue (1911) with *Ophiotaenia adiposa* Rudin, 1917 as its type taxon (<https://www.gbif.org/species/4412478>)¹¹.

Ophiotaenia tigrina (Woodland, 1925) is an omnipresent species freshwater fishes, amphibians and reptiles¹²⁻¹⁴. *O. tigrina* belongs to family proteocephalidae. These parasites are thin, small and milky white in color measuring 12-14 cms in length, acraspedote and apolytic. Extensive breakthrough has been made on the phylogenetic studies of tapeworms (Euscestoda) and rDNA clusters are proved to be effective tools for genetic studies¹⁵. In this study, 28S ribosomal DNA of cestode *O. tigrina* was used. This ribosomal DNA was proved to be an excellent tool in resolving many taxonomic issues of helminth parasites¹⁵⁻¹⁹. A section of the large nuclear subunits of 28S rDNA gene was chosen in determining the interspecific phylogenetic relationships with other cestode groups^{6,20,21}. This molecular marker was extensively used for taxonomic studies as it is easy to amplify even small quantities of parasitic DNA. The

present study is designed to identify this cestode based on morphological studies supplemented with molecular characterization.

2. MATERIALS AND METHODS

2.1. Study area

Three hundred adult frogs, *Hoplobatrachus tigerinus* were collected with the aid of butterfly nets or by hand from four different sites of YSR Kadapa District (Lat. 14°28'N 78°49'E, 137 m Altitude), located in Andhra Pradesh state during February, 2013 to February, 2015. The collection sites include (Figure.1 & 2):

Site 1: Industrial Estate area (Lat. 14°47'N 78°76'E, 138 meters Altitude), YSR Kadapa district,

Site 2: Campus area of Yogi Vemana University (Lat.14°28'N 78°49'E, 137 m Altitude), located in YSR Kadapa District of Andhra Pradesh,

Site 3: Ramapuram village (14.05°N 78.75°E, 143 meters) Raychoti Mandal,

Site-4: Bouinpalli village, Kadapa (Lat.14°28'N 78°52'E, 379 meters).

2.2. Ethical Considerations

All procedures contributing to this work fulfill with the ethical values of the appropriate national guides on the care and use of laboratory animals and have been approved and authorized by IAEC (Institution of Animal Ethics Committee-Regd. No.1460/PO/a/11/CPCSEA, dt. 20.05.2011), Zoology Department in Faculty of Life Sciences, Yogi Vemana University, Andhra Pradesh.

2.3. Parasite Collection

Host samples of different sizes i.e., small, medium and large were transported to laboratory and were instantly scrutinized after being euthanized. The intestinal region was deviscerized and its contents were scraped into a petridish containing 0.7% solution or Ringer's solution to examine for helminths under a stereozoom microscope (LM-52-3621 Elegant). These helminth parasites were found attached to intestine of the host with their scolex. The acquired parasites were alienated into three groups for the subsequent analyses.

2.4. Morphological Analysis

Collected specimens were temporarily preserved in an ideal fixative, A.F.A (Alcohol-85%, Formalin- 5%, Glacial acetic acid- 10%) for 24 hours for the whole mount preparations and stained with alum carmine

followed by proper dehydration with a graded series of alcohols (70%, 80%, 95%, 100%), xylene clearing and lastly mounted on a glass slide with Canada balsam or DPX mount^{22,23}. Line diagrams were drawn with the aid of drawing tube accessory of Lynx Trinocular microscope (N-800M). Morphological information was noted from the whole mount and ocular micrometer dimensions were given in millimeters unless otherwise stated. Identifications were made according to Yamaguti (1959)²⁴.

2.5. Scanning Electron Microscopy (SEM) study

For SEM analysis, the parasites were fixed in 2.5% glutaraldehyde in 0.1 M phosphate buffer (pH 7.2) at 4°C and processed according to the standard protocols described²⁵. SEM photographs at various magnifications were captured with a Carl Zeiss Scanning electron microscope (SIGMA TM) facility at Sri Venkateswara University, Tirupati.

2.6. Molecular Characterisation

A portion of collected parasites were fixed in 95% ethyl alcohol (molecular grade) for molecular analysis and placed at low temperatures of -20°C until DNA extraction. The genomic DNA of the parasite was extracted from the scolex, mid-section of parasites and terminal proglottids following the standard Phenol-Chloroform procedure^{16,26}. A portion of large nuclear subunit of 28S ribosomal DNA was selected as a molecular marker due to an easy amplification with very minute quantity of DNA (Figure 3 & 4). The genomic DNA of two cestode parasites, *Paraorygmatobothrium floraforme* from *Rhizoprionodon acutus* (Shark)²⁷ and *Ophiotaenia tigrina* from *Hoplobatrachus tigerinus* (Indian bull frog) were isolated for molecular characterization from this Parasitology laboratory division, Department of Zoology, Yogi Vemana University, Kadapa.

The subsequent PCR primers i.e.,

1) Forward primer LSU5 (5'-TAGGTCGACCCGCTGAAYTTAAGCA-3')
 2) Reverse primers 1200R (5'-GCATAGTTCACCATCTTTTCGG-3') and
 3) 1500R (5'-GCTATCCTG AGGGA AACTTCG-3') were attained from earlier available studies and these PCR products were amplified with AURA Red TaqX master mix and DNA easy Tissue kit (Qiagen) was used to purify the amplified products. Cycling conditions were sustained in following steps: 1) Initial denaturation for 1 minute at 95 °C for 5 minutes, denaturation at 95 °C (30 seconds), annealing at 52°C (30 seconds) and extension at 72°C (30 seconds). Step-2: 40 cycles, final extension at 72°C for 7 minutes. An automated sequencer (Eurofins, Chennai) was used to purify and sequence the PCR products in both directions and the

obtained sequences were submitted to the NCBI gene bank get the accession number (KY574290). The sequences generated in this study were compared with available sequences in the genomic databases using bioinformatic tools such as Basic Local Alignment Search Tools (BLAST) on NCBI (<http://www.ncbi.nih.gov>). The dissimilarities were compared by executing the pairwise alignment of the nucleotides. The phylogenetic tree of this cestode was depicted using MEGA 6.0 software and Test Neighbor-Joining (NJ) analysis was performed with a branch support of using 500 bootstrap replicates.

3. RESULTS

3.1. Morphological Analysis

Ophiotaenia tigrina (Woodland, 1925) Gupta and Arora, 1979

Synonyms: *Proteocephalus tigrinus* Woodland, 1925

Name of the host: *Hoplobatrachus tigerinus*

Number of hosts examined: 300

Number of hosts infected: 18

Number of parasites collected: 31

Site of infection: Intestine

Prevalence of infection: 10.33%, Mean intensity:

1.72, Mean abundance: 0.103.

Description: Morphological assessment of the parasite revealed distinctive proteocephalid characteristics: Parasites small, thin and milky white in colour measuring 12-14 cm in length. Strobila acraspedote. Scolex small, 1.15-1.20 in length and 0.38-0.41 in width with four uniloculate cup shaped suckers (Figure. 5 & 6). Strobila consisting of three types of proglottids: Immature proglottids 0.18-0.22 × 0.42-0.47, mature proglottid 0.22-0.32 × 0.48-0.76 and gravid proglottids 0.58-0.66 × 1.76-1.90 (Figure. 7 & 8). Testes medullary, 30-50 in number, spherical 0.04-0.09 occupies entire proglottid on either side of ovary. Cirrus sac extended to anterior half 0.48-0.55 × 0.02-0.028. Ovary bilobed, 0.25-0.29 × 0.26-0.28. Ovarian lobes are connected by an isthmus and extend up to the level of cirrus sac. All these morphological characters are in accordance with those of *Ophiotaenia tigrina* species.

3.2. SEM description

The SEM (Figure.9) studies of the scolex shows four uniloculate cup shaped suckers with a spiny rostellum at the tip. Neck is long and strobila consists of proglottids (Figure.10).

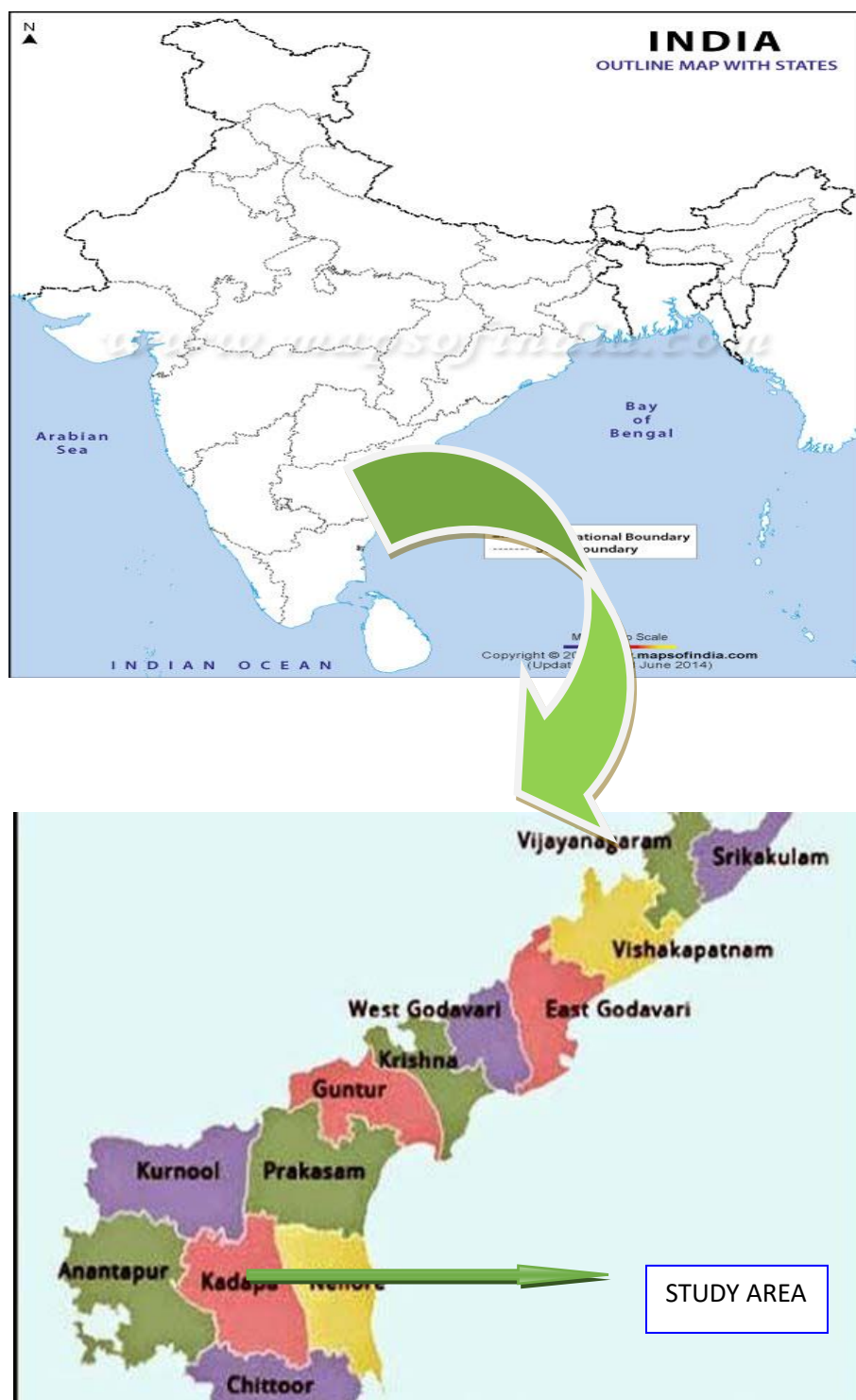
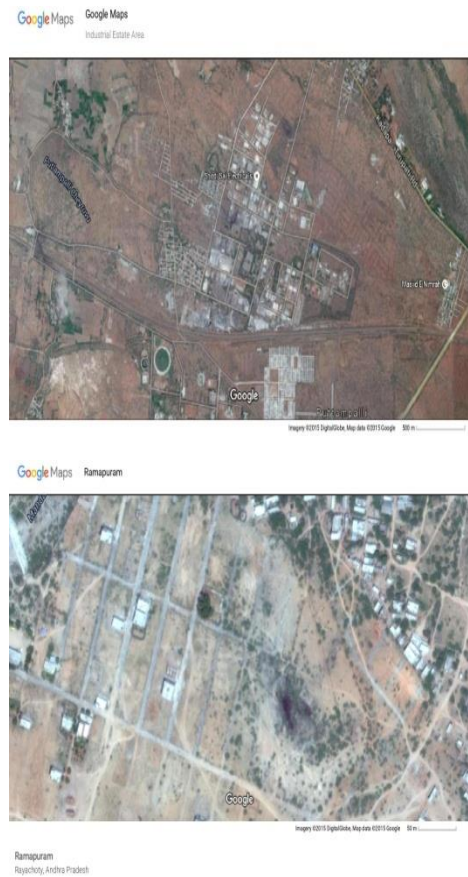


Figure.1: India map showing study area- Kadapa District in Andhra Pradesh

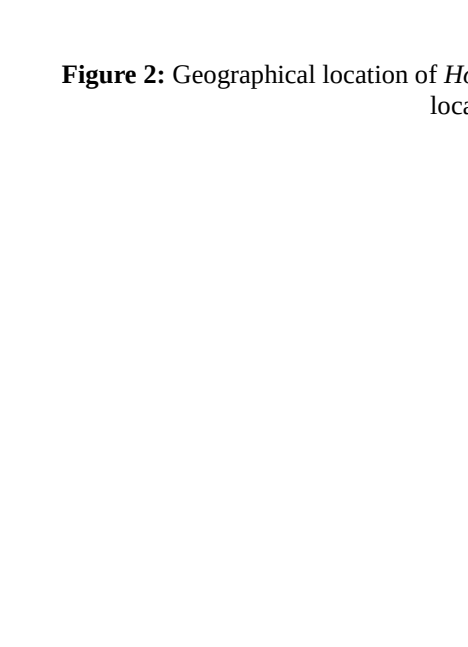
Site 1- Industrial Estate Area



Site 2- Yogi Vemana University Campus



Site 3 - Ramapuram village, Rayachoti



Site 4- Bouinipalli Village



Figure 2: Geographical location of *Hoplobatrachus tigerinus* and *Duttaphrynus melanostictus* collected from 4 locations of YSR Kadapa, Andhra Pradesh

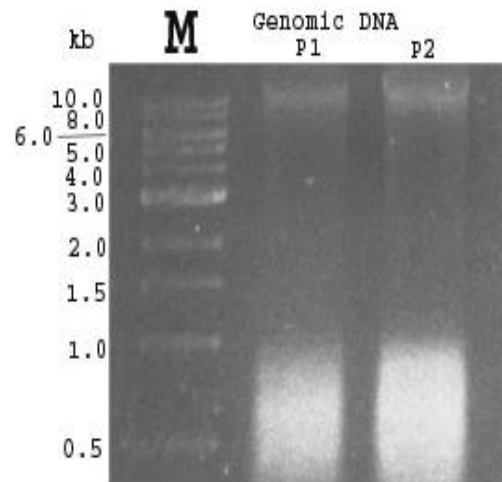


Figure.3: Genomic DNA isolation (M-DNA marker (100bp ladder), P1-DNA of *Paraorygmatobothrium floraforme* from Shark, *Rhizoprionodon acutus* & P2- *Ophiotaenia tigrina* from *Hoplobatrachus tigerinus* DNA of different cestode parasites)

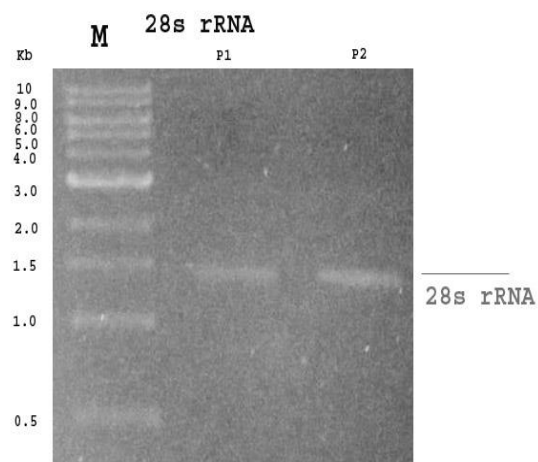


Figure 4: 28S rRNA isolation (M-DNA marker (100bp ladder), P1 & P2 28sRNA of these cestode parasites)



Figure 5: Microphotograph of Scolex- *Ophiotaenia tigrina* (0.2mm)

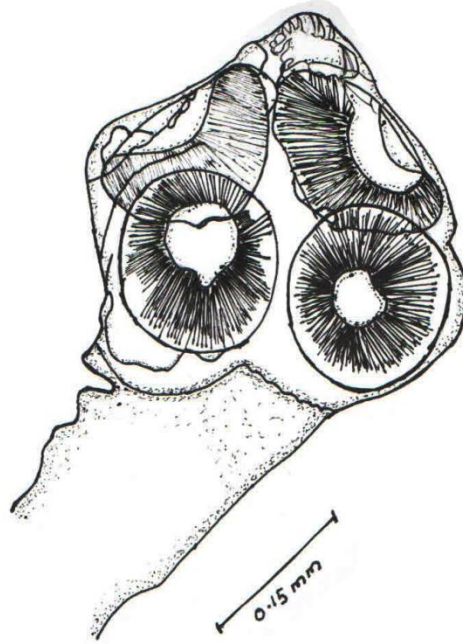


Figure 6: Line Diagram of scolex- (0.15mm)

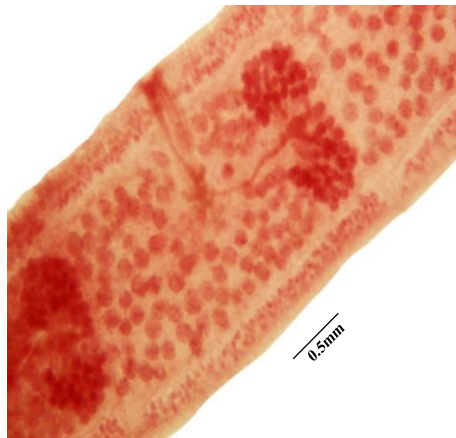


Figure 7: Microphotograph of mature proglottid- *Ophiotaenia tigrina* (0.5mm)

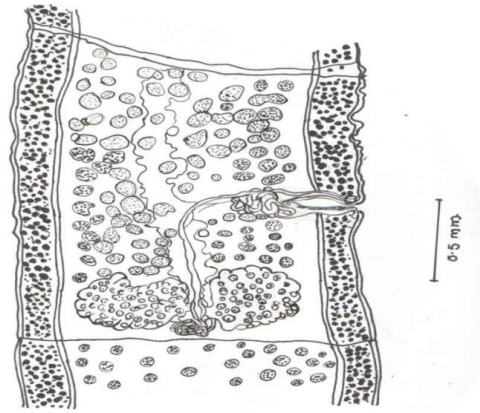


Figure 8: Line diagram of mature proglottid- *Ophiotaenia tigrina*- (0.5mm)

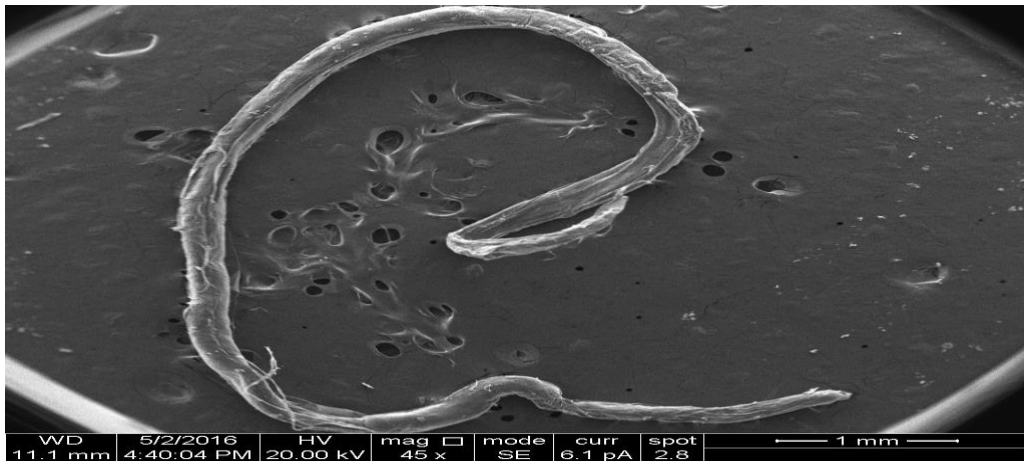


Figure 9: SEM photograph of *Ophiotaenia tigrina*, whole worm-1mm



Figure 8: SEM photograph of scolex of *Ophiotaenia tigrina*, 50µm

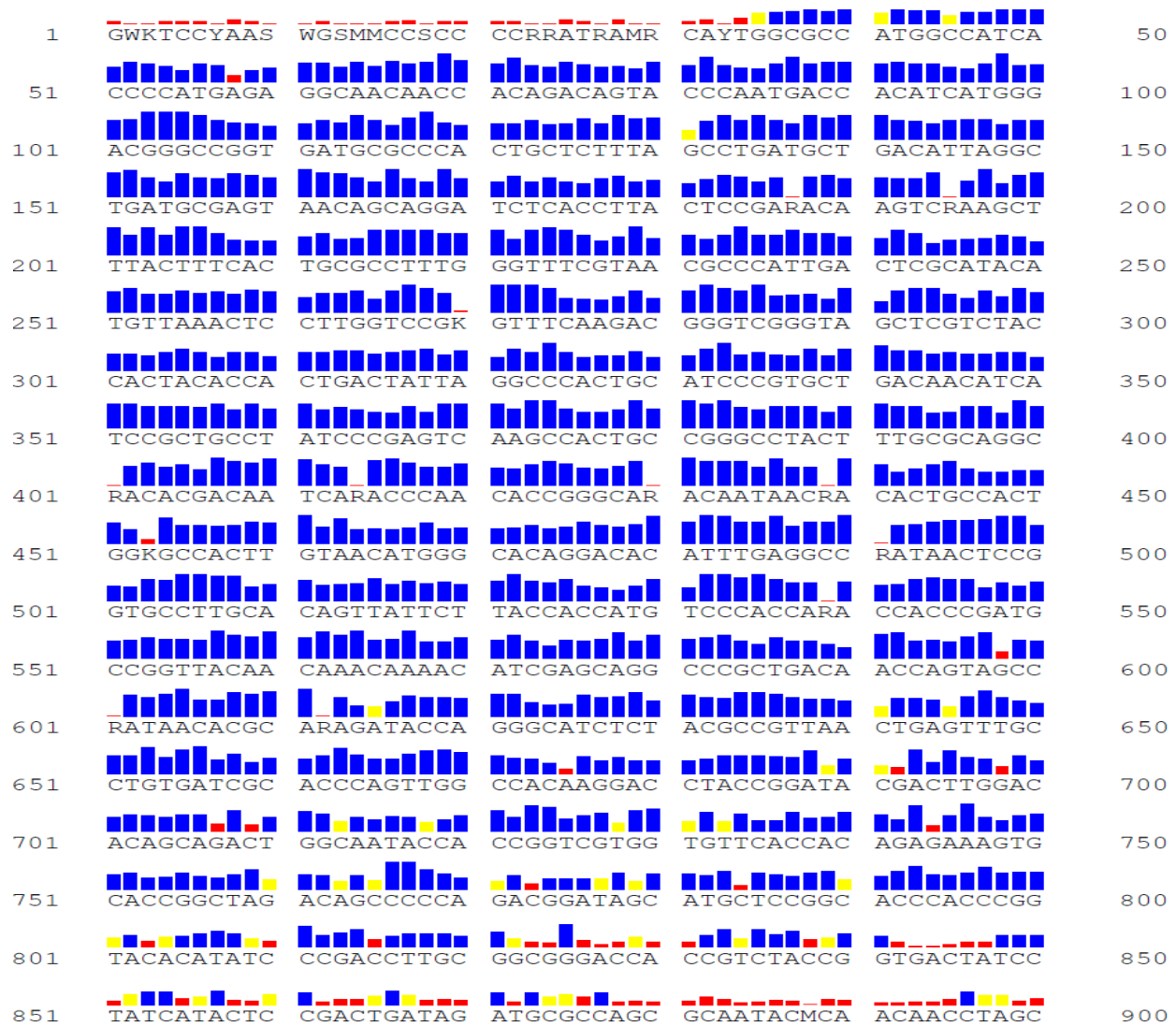


Figure 11: Forward Nucleotide Sequence of rDNA of *O. tigrina*

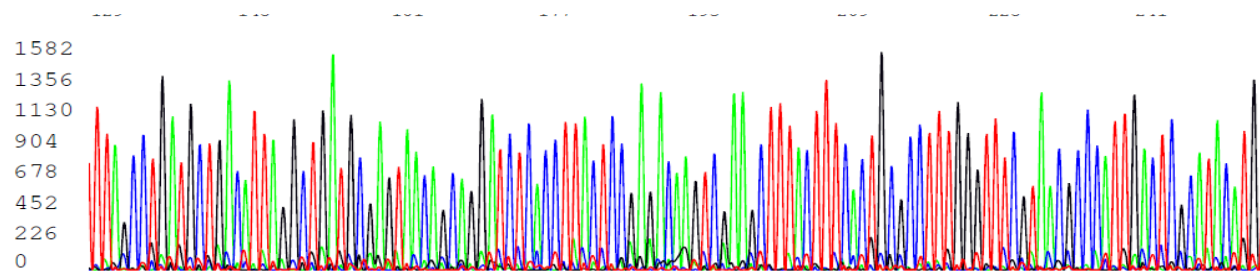
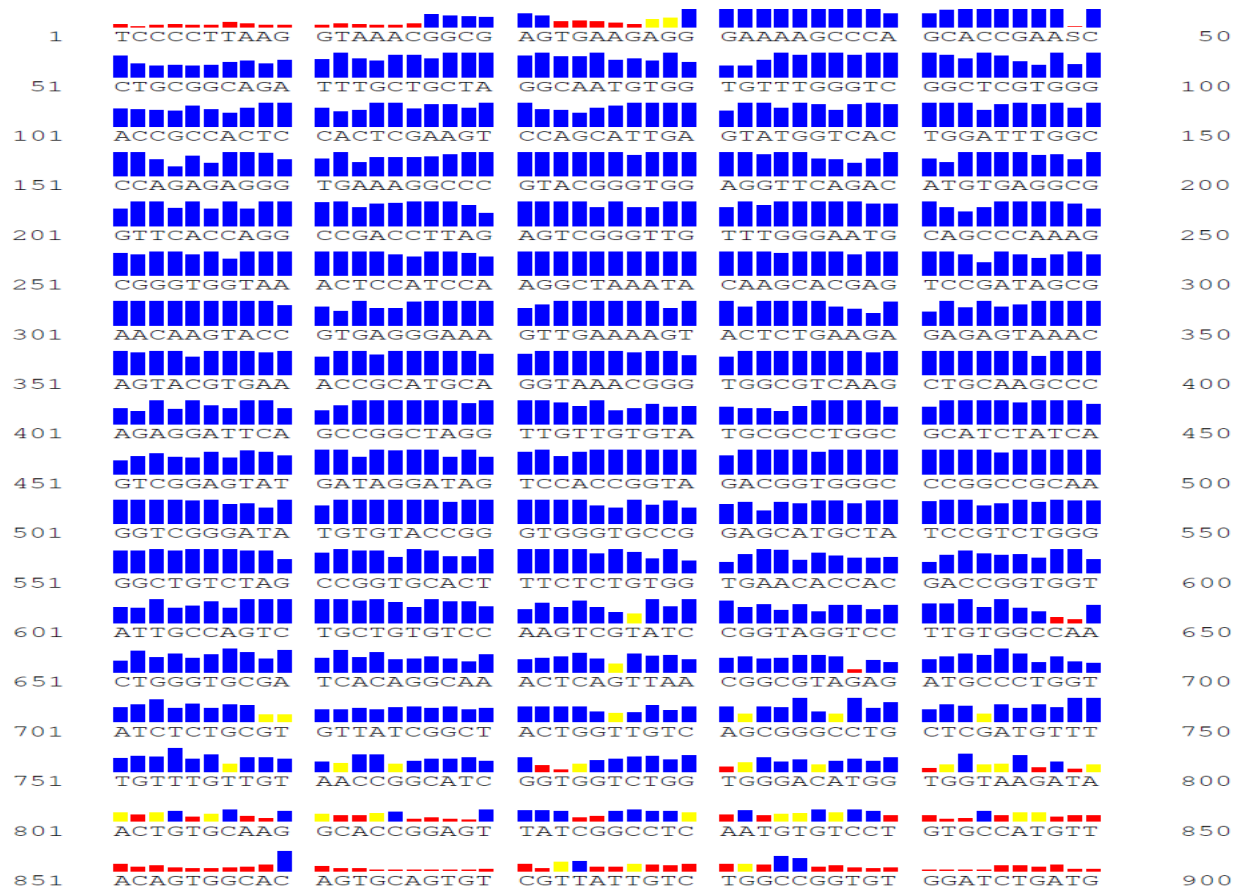


Figure 12: Partial sequence of the parasite *O. tigrina*



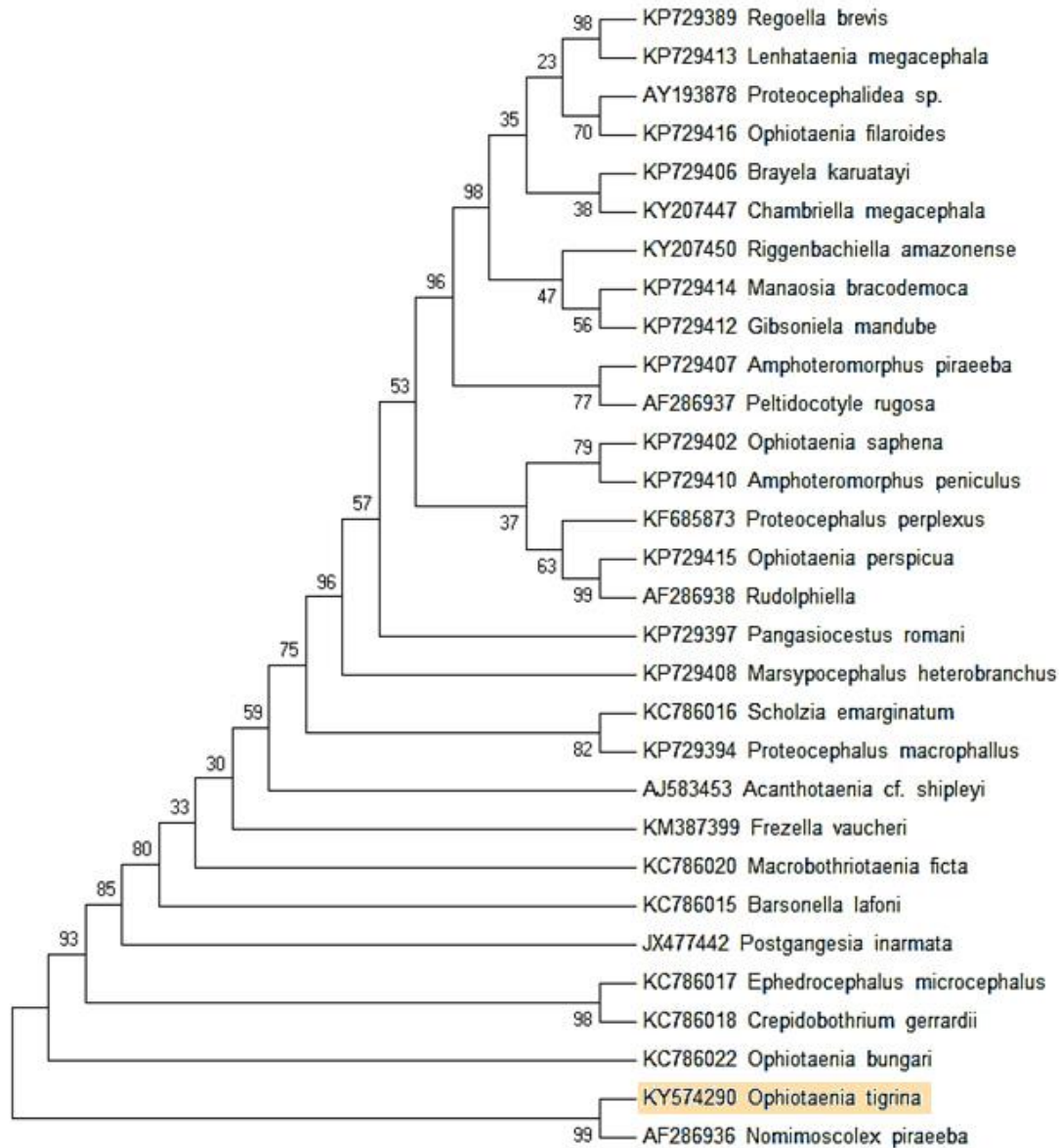


Figure 15: Tree-1-Showing phylogenetic relationship of present species with Proteocephalid cestodes of different geographical areas collected from Genbank

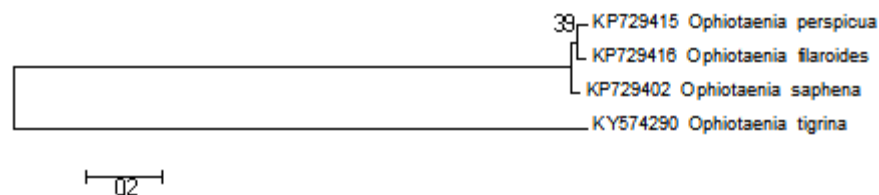


Figure 16: Tree-2-Showing phylogenetic relationship of present species with available sequences of *Ophiotaenia* cestodes from different geographical area

Molecular analysis:

DNA sequence length of the 28 rDNA of *O. tigrina* ranged from 937 to 961 bp with forward, reverse and partial nucleotide sequences (Figure.11-14). The nucleotide sequences data reported in this paper were submitted to the gene bank by the name *Ophiotaenia tigrina* with an accession number (KY574290). These sequences were subjected to NCBI blast. The BLAST hit results shows that the sequences of the Order Proteocephalidea and family Proteocephalidae species are closer to this sequence with maximum similarity.

Phylogenetic analysis:

For the purpose of phylogentic analyses, molecular data was produced for one specimen. The phylogenetic tree was generated based on the various cestode parasites isolated from the different vertebrates of various topographical areas. MEGA 6 software was used to align all 28S ribosomal DNA sequences. The obtained nucleotide sequences from PCR were placed to NCBI BLAST and evaluated with other available cestode sequences from the Gene Bank. The BLAST hits results clearly depicted the close resemblances of the present cestode sequences with the sequences of the cestodes of the family Proteocephalidae, thus confirming their inclusion into this family. The two phylogenetic trees were drawn based on the available nucleotide sequences in Gene bank i.e., one tree was drawn by comparing the present species nucleotide sequence with the sequences of proteocephalid cestodes of different vertebrates from different geographical isolates, while the other tree was drawn by comparing the present species sequences with the only the available *ophiotaenia* cestode parasites sequences in genebank. Phylogenetic tree-1 showed maximum close similarities (99%) of the proteocephalid cestode sequences with the present sequence confirming its position to this family (Figure.15). While the Phylogenetic tree (Tree 2) showed that the present species sequences matched only 39% with the sequences other ophiotaenid cestodes from different geographical isolates confirming it as *Ophiotaenia tigrina* (Figure.16).

4. DISCUSSION

The present tapeworm was assigned to the genus *Ophiotaenia* of the order Proteocephalidea Mola, 1928 based on the basic characteristics of the genus. The cestodes of the order Proteocephalidea Mola, 1928 are parasites of freshwater fishes, amphibians, reptiles and marsupials^{3,4,7}, in contrast, amphibians are scarcely represented as the definitive hosts of proteocephalidean cestodes^{8,9}. The genus

Ophiotaenia is one among the 19 genera within the subfamily Proteocephalinae with nearly 96 currently recognized species parasitizing amphibians, reptiles and birds of which, 25 species among them parasitize amphibians exclusively^{5,7-10,14}. Brooks (1978) discussed the status of *Ophiotaenia* and argued with the proposals made by Rudin (1917) and Freze (1965) and considered *Ophiotaenia* as a junior synonym of *Proteocephalus*^{2,28,29}. However, Chambrier *et al.*, (2004) and Ammann and Chambrier (2008) evaluated the relative ovarian sizes of the species of the genus *Proteocephalus* and *Ophiotaenia* and discriminated them as separate genera under the proteocephalinae subfamily^{15,30}. Hence, the present specimens are placed under the genus *Ophiotaenia*. From India, there are only 2 species reported from *H. tigerinus* i.e., *O. junglensis* (Srivastava and Capoor, 1980) and *O. tigrina* (Woodland, 1925) Gupta and Arora, 1979. But, *O. junglensis* is considered as *species inquirenda* according to IZCN. Hence, the present tapeworm was given the status as *Ophiotaenia tigrina* based on the medullary distribution of gonads as well as the presence of four simple unilocular suckers and two testis fields^{2,7}. The present study represents a small molecular phylogenetic analysis of the frog tapeworms of family *Proteocephalidae* and these types of studies are the need of the hour to resolve many taxonomic issues in helminth identification.

5. CONCLUSION

The wild borne frogs are known to be definitive and intermediate hosts to a variety of metazoan parasite as they inhabit each and every organ. Intestine is one such organ which is very heavily infected with various types of metazoan parasites. Species identification through morphological analysis is very difficult and needs the accuracy in identification between very closely related species, which can be achieved through molecular characterization. This study presents a molecular phylogenetic analysis of the frog tapeworms of family *Proteocephalidae* due to their elongated body suitable enough for DNA extraction and further processing. This is the first examination of molecular analysis of cestode parasites from anurans from the YSR Kadapa district, Andhra Pradesh. DNA sequence length ranged from 937 to 961 bp and the two phylogenetic trees were drawn based on the sequences collected from Gene bank. These types of analyses can reveal the interrelationship between different genera within a single family.

Conflicts of Interest

There are no conflicts of interest.

Acknowledgement

The First author, Hemalatha Mannela is grateful to University Grants Commission for providing the financial assistance as JRF and SRF under Rajiv Gandhi National fellowship (RGNF Award No.2011-12/RGNF-SC-AND-5015, dt.01.06.2012).

References

- Olson, P.D., D.T.J. Littlewood, R.A. Bray and J. Mariaux, 2001. Interrelationships and evolution of the tapeworms (Platyhelminthes: Cestoda). *Molecular Phylogenetics and Evolution*, 18: 443–467.
- Freze, V.I., 1965. [Proteocephalata. Tapeworms of Fish, Amphibians and Reptiles]. In: SkryabinKI (Ed.) [Principles of Cestodology. Vol. 5]. Izdatel'stvo «Nauka», Moscow, 538 pp. [In Russian] [English translation, Israel Program of Scientific Translation, 1969, Cat. No. 1853. v + 597 pp.].
- Rego, A.A., 1994. Order Proteocephalidea Mola, 1928 In: Khalil LF, Jones A, Bray RA (Eds) *Keys to the cestode parasites of vertebrates*. CAB International, Wallingford, 257–293.
- Cañeda-Guzmán, I.C., A. de Chambrier and T. Scholz, 2001. *Thaumasioscolex didelphidis* n. gen., n. sp (Eucestoda: Proteocephalidae) from the black-eared opossum *Didelphis marsupialis* from Mexico, the first proteocephalidean tapeworm from a mammal. *Journal of Parasitology*, 87: 639–646.
- De Chambrier, A., T.T. Binh and T. Scholz, 2012. *Ophiotaenia bungari* n. sp. (Cestoda), a parasite of *Bungarus fasciatus* (Schneider) (Ophidia: Elapidae) from Vietnam, with comments on relative ovarian size as a new and potentially useful diagnostic character for proteocephalidean tapeworms. *Systematic Parasitology*, 81: 39–50.
- De Chambrier, A., A. Waeschenbach, M. Fisseha, T. Scholz and J. Mariaux, 2015. A large 28S rDNA-based phylogeny confirms the limitations of established morphological characters for classification of proteocephalidean tapeworms (Platyhelminthes, Cestoda). *ZooKeys*, 500: 25–59.
- Schmidt, G.D., 1986. *Handbook of Tapeworm identification*. CRC Press, Boca Raton, 675pp.
- de Chambrier, A., S.C. Coquille and D.R. Brooks, 2006. *Ophiotaenia bonneti* sp. n. (Eucestoda: Proteocephalidea), a parasite of *Rana vaillanti* (Anura: Ranidae) in Costa Rica. *Folia Parasitologica*, 53: 125–133.
- Marsella, C.M.V. and A. de Chambrier, 2008. *Ophiotaenia alessandrae* sp. n. (Eucestoda: Proteocephalidea), a parasite of *Hyla boans* (Anura: Hylidae) from Amazonia in Ecuador. *Revue Suisse de Zoologie*, 115: 553–563.
- De Chambrier, S., A. de Chambrier, 2010. Two new genera and two new species of proteocephalidean tapeworms (Eucestoda) from reptiles and amphibians in Australia. *Folia Parasitologica*, 57: 263–279.
- La Rue, G.R., 1911. A revision of the cestode family Proteocephalidae. *Zoologischer Anzeiger*, 38: 473–482.
- Woodland, W.N.F., 1925. On three new proteocephalids (Cestoda) and a revision of the genera of the family. *Parasitology*, 17: 370–394.
- Gupta, N.K. and Arora, S., 1979. On *Proteocephalus tigrinus* Woodland, 1925 (Proteocephaloidea: Proteocephalidae), a cestode parasite of *Rana tigrina* at Amritsar (Punjab, India). *Res. Bull. (Science) Punjab University*, 30: 79–82.
- De Chambrier, A., and A. Gil de Perterra, 2012. *Ophiotaenia oumanskyi* sp.n. (Eucestoda: Proteocephalidea), a parasite of *Lepidobatrachus laevis* Budgett, 1899 (Anura: Leptodactylidae) from Paraguay. *Revue Suisse de Zoologie*, 119 (4): 561–570.
- De Chambrier, A., M. Zehnhnder, C. Vauchcher and J. Mariaux, 2004. The evolution of the Proteocephalidea (Platyhelminthes, Eucestoda) based on an enlarged molecular phylogeny, with comments on their uterine development. *Systematic Parasitology*, 57: 159–171.
- Hillis, D.M. and M.T. Dixon, 1991. Ribosomal DNA: Molecular evolution and phylogenetic inference. *Quarterly Review of Biology*, 66: 411 (PMID: 1784710).
- Zehnder, M.P. and J. Mariaux, 1999. Molecular systematic analysis of the order Proteocephalidea (Eucestoda) based on mitochondrial and nuclear rDNA sequences. *International Journal for Parasitology*, 29: 1841–1852.

18. Hypša, V., A. Škeříková and T. Scholz, 2005. Phylogeny, evolution and host-parasite relationships of the order Proteocephalidea (Eucestoda) as revealed by combined analysis and secondary structure characters. *Parasitology*, 130: 359–371.
19. Waeschenbach, A., B.L. Webster, D.T. Littlewood, 2012. Adding resolution to ordinal level relationships of tapeworms (Platyhelminthes: Cestoda) with large fragments of mtDNA. *Molecular Phylogenetics and Evolution*, 63: 834–847.
20. Malsawmtluangi, C., K.P. Prasad, K.D. Biswal and V. Tandon, 2011. Morphological and molecular identification of the metacestode parasitizing the liver of rodent hosts in bamboo growing areas of Mizoram, North east India. *Bioinformation*, 7(8): 393-399.
21. Justine, J.L., M. Briand and A.R. Bray, 2012. A quick and simple method, usable in the field, for collecting parasites in suitable condition for both morphological and molecular studies. *Parasitology Research*, (in press).
22. Hiware, C.J., B.V. Jadhav and A.D. Mohekar, 2003. *Applied Parasitology. A practical manual*. Mangaldeep Publications.
23. Madhavi, R., C. Vijayalakshmi and K. Shyamasundari, 2007. *Collection, staining and identification of different helminth parasites*. A manual of the workshop on fish parasites- Taxonomy Capacity Building, Andhra University Press.
24. Yamaguti, S., 1959. *Systema helminthum*, Vol-II. The cestodes of vertebrates. Interscience Publishers Inc., New York, New York. 699pp.
25. Elizabeth, R.F., T.H. Bryan, Vinod Nair, H.H. Forest and W.D. David, 2012. Scanning electron Microscopy. *Current Protocols in Microbiology*, Chapter 2. B.2.
26. Sambrook, J., Fritsch, E.F., and T. Maniatis, 1989. *Molecular cloning. A laboratory manual* (second edition). Cold Spring Harbor Laboratory Press, New York.
27. Srinivasa Kalyan C and V. Anu Prasanna, 2018. Morphological and Molecular Identification of *Paraorygmato bothrium floraformis* in *Rhizoprionodon acutus* off Nellore Coast, Bay of Bengal, India. *Trends in Bioinformatics*, 11: 25-32.
28. Brooks, D.R., 1978. Evolutionary history of the cestode order Proteocephalidea. *Systematic Zoology*, 27: 312–323.
29. Rudin, E., 1917. Ichthyotaenien der Reptilien. *Rev. Suisse Zool.*, 25: 14-381.
30. Ammann, M., and A. de Chambrier, 2008. *Ophiotaenia gilberti* sp. n. (Eucestoda: Proteocephalidea), a parasite of *Thamnodynastes pallidus* (Serpentes: Colubridae) from Paraguay. *Revue Suisse de Zoologie*, 115: 565–573.